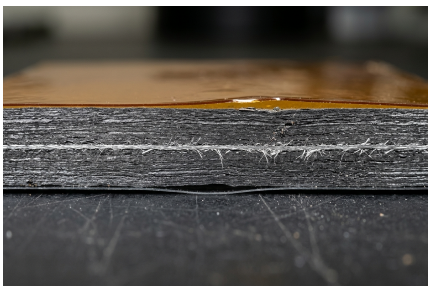
GRAPHITE-FLEX SERIES (GFlex-1000 / GFlex-1500)

# InsulTherm Graphite-Flex™

## 5-Layer Edge-Sealed Pyrolytic Graphite Lateral Heat Spreader (ASTM E595 Low-Outgassing)

InsulTherm Graphite-Flex™ is an ultra-thin, 5-layer carrier-backed synthetic pyrolytic graphite laminate engineered to eliminate intense localized processor, GaN/RF, and optical hotspots by spreading heat laterally (1,000–1,500 W/m·K in-plane—up to 4× pure copper at <25% weight). Integrating a neutral-axis woven E-Glass scrim, an edge-overlapping 0.025 mm Kapton® dielectric cap (>5 kV), and an ASTM E595 silicone-free adhesive, it prevents carbon flaking and optical lens fogging.



Macro Cross-Section / Studio Hardware Inspection

| 1,000–1,500 W/m·K | ASTM E595 (TML <1%) | >5 kV Kapton Cap | Silicone-Free | GTE Ban Compliant |
|---|---|---|---|---|

## 1. CHEMICAL / MATERIAL COMPOSITION & INTERNAL LAYER ARCHITECTURE

Polymeric precursor film carbonized above 2,500°C into oriented hexagonal SP2 carbon planes. Dual pyrolytic graphite cores are laminated symmetrically across an ultra-thin woven E-Glass fiberglass neutral-axis scrim using a 100% siloxane-free ASTM E595 acrylic PSA (TML < 1.0%, CVCM < 0.1%) beneath a 0.025 mm translucent amber Kapton® polyimide dielectric shield.

| | |
|---|---|
| **Layer 1 (0.025 mm)** | Translucent Amber Kapton® Polyimide Dielectric Top Cap (Edge-Sealed, >5 kV) |
| **Layer 2 (Upper Core)** | High-Crystallinity Oriented Pyrolytic Graphite Matrix (k_xy = 1,000–1,500 W/m·K) |
| **Layer 3 (Center Axis)** | Embedded Woven E-Glass Fiberglass Scrim + ASTM E595 Bonding Line |
| **Layer 4 (Lower Core)** | Balanced Lower Synthetic Pyrolytic Graphite Spreader Matrix |
| **Layer 5 (0.050 mm)** | Clear Siliconized PET Release Liner + 0.010 mm Silicone-Free Acrylic PSA |

## 2. TYPICAL PHYSICAL, THERMAL & ELECTRICAL PROPERTIES

| PROPERTY / PARAMETER | TYPICAL ENGINEERING VALUE | TEST METHOD / STANDARD |
|---|---|---|
| **In-Plane Conductivity (X-Y)** | 1,000 – 1,500 W/m·K (4× Copper diffusivity) | Laser Flash (ASTM E1461) |
| **Through-Plane Conductivity (Z)** | 7.0 – 15.0 W/m·K | ASTM E1461 |
| **Vacuum Outgassing (No Silicone)** | TML < 1.0% \| CVCM < 0.10% (Zero oil fogging) | ASTM E595 (125°C, 10^-5 Torr) |
| **Top-Cap Dielectric Isolation** | > 5,000 V AC breakdown (0.025 mm Kapton®) | ASTM D149 |
| **Composite Total Thickness** | 0.15 mm – 0.35 mm ultra-low-profile stack | ASTM D374 |
| **Graphite Core Density** | 1.95 – 2.12 g/cm³ (~76% lighter than Copper) | Helium Pycnometry |
| **Operating Temperature Range** | -55°C to +150°C (PSA) / +400°C (Core) | MIL-STD-810H Shock |
| **Minimum Bend Radius** | 2.0 mm dynamic / 1.0 mm static routing | IPC-TM-650 Flex |
| **Flammability Rating** | UL 94 V-0 | UL 94 |
| **Procurement Sovereignty** | 100% India DVA (< ₹200 Cr MoD GTE Compliant) | Make-in-India Class I/II |



InsulTherm Innovations, Advanced Materials & Digital Converting Wing, Block C-14, Sanand GIDC Industrial Estate (Near iCreate Campus), Ahmedabad, Gujarat – 382110, India

Engineering Notice: Typical laboratory & baseline metrics (ASTM D5470 / UL 94 / ASTM E595). System-level OEM qualification is required prior to volume release.

## 3. ORDERING NOMENCLATURE, PREMIX VARIANTS & COMMERCIAL PRICING SCHEDULE

| PART / GRADE CODE | MATERIAL SPECIFICATION & CONSTRUCTION | PROTOTYPE / R&D TIER | VOLUME / OEM TIER |
|---|---|---|---|
| `GFlex-1000 (Area-Rated)` | Kapton® Cap + Graphite (1,000 W/m·K) + Scrim: ≤25cm² ₹350 \| +₹5.50/cm² | **₹350–₹1,800 / pc (by cm²)** | **₹3.20 / cm² nested (min ₹28)** |
| `GFlex-1500 (Area-Rated)` | Satellite/Radar Grade (1,500 W/m·K): ≤25cm² ₹550 \| +₹7.50/cm² up to 300cm² | **₹550–₹1,800 / pc (by cm²)** | **₹4.60 / cm² nested (min ₹38)** |
| `Space/Defense Kit` | Turnkey Zero-Outgassing Sampler Kit (GFlex + Ultra-NS + ASTM E595 Trace) | **₹12,500 / turnkey lab kit** | **Ships in 48 Hours** |

## 4. TARGET INDUSTRIAL APPLICATIONS & DEPLOYMENT SECTORS

- Defense UAV flight computers, phased-array radar Transmit/Receive (T/R) modules, and seeker electronics (BEL, HAL, DRDO, private primes).
- Low-Earth-Orbit (LEO) satellite payloads and optical Earth-observation camera housings requiring zero siloxane outgassing.
- Sealed automotive ADAS LiDAR/camera ECUs and 5G/6G small-cell telecom enclosures with zero vertical clearance for finned heatsinks.
- 800G / 1.6T pluggable optical transceivers in AI data centers routing heat laterally into cold-rail chassis walls.

## 5. DIGITAL CNC CONVERTING, APPLICATION & STORAGE GUIDELINES

1. Edge-Seal Integrity: Specify perimeter geometry in .DXF CAD so InsulTherm's CNC step-cut creates a 1.0–1.5 mm amber Kapton® dielectric skirt overhang around conductive graphite edges where adjacent SMD components sit within <2 mm. 2. Application: Peel the clear 0.050 mm PET release liner using the kiss-cut tab and apply with 15–30 psi roller pressure onto clean IC/metal surfaces. 3. Cleanroom Option: Available double-bagged in N2-purged ESD pouches for SpaceTech and optical assembly lines.

## 6. QUALITY ASSURANCE, REGULATORY COMPLIANCE & SOVEREIGN DVA TRACEABILITY

| | |
|---|---|
| **Thermal & Electrical Metrology** | Batch-wise ASTM D5470-17 thermal impedance calibration curves (10–50 psi) & ASTM D149 dielectric verification. |
| **Flammability & Fire Safety** | UL 94 V-0 rated material formulations; supports ARAI AIS-156 Amendment 3 & UL 9540A pack/rack propagation testing. |
| **Outgassing & Environmental** | RoHS 3 (EU 2015/863) & REACH compliant; ASTM E595 low-outgassing trace on Graphite-Flex™ & Ultra-NS™ grades. |
| **Domestic Value Addition (DVA)** | 100% India Digital CNC Converting (Gujarat); compliant with PM E-DRIVE, SECI BESS DVA & <₹200 Cr MoD GTE Ban. |

**RAPID 24–48 HOUR CAD-TO-PART PROTOTYPING & OEM PROCUREMENT DESK**
Submit .DXF / .STEP drawings, request <₹15,000 Petty-Cash R&D Sampler Kits, or schedule an engineering review:
**Primary Official Desk:** manav.sb@inslthrm.in | **Active Operations & Order Routing:** manavsb9719@gmail.com
**Engineering Helpline:** +91 78148 XXXXX | **Manufacturing Wing:** InsulTherm Innovations, Advanced Materials & Digital Converting Wing, Block C-14, Sanand GIDC Industrial Estate (Near iCreate Campus), Ahmedabad, Gujarat – 382110, India